

```
[*]  
...  
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 147536 entries, 0 to 147535  
Data columns (total 14 columns):
```

#	Column	Non-Null Count	Dtype
0	title	147536 non-null	object
1	country	147406 non-null	object
2	location	147405 non-null	object
3	building_construction_year	64719 non-null	float64
4	building_total_floors	68224 non-null	float64
5	apartment_floor	54592 non-null	float64
6	apartment_rooms	74178 non-null	float64
7	apartment_bedrooms	36982 non-null	float64
8	apartment_bathrooms	55973 non-null	float64
9	apartment_total_area	141796 non-null	object
10	apartment_living_area	27712 non-null	object
11	price_in_USD	144961 non-null	float64
12	image	147536 non-null	object
13	url	147536 non-null	object

```
dtypes: float64(7), object(7)  
memory usage: 15.8+ MB
```

	building_construction_year	building_total_floors	apartment_floor	apartment_rooms	apartment_bedrooms	apartment_bathrooms	price_in_USD
count	64719.000000	68224.000000	54592.000000	74178.000000	36982.000000	55973.000000	1.449610e+05
mean	1996.921754	8.575692	5.791709	2.572097	2.289222	1.364229	4.121722e+05
std	157.527635	8.356781	5.541368	1.319545	18.276913	0.745019	8.420984e+05
min	1.000000	-1.000000	-2.000000	-1.000000	-1.000000	1.000000	0.000000e+00
25%	2004.000000	2.000000	2.000000	2.000000	1.000000	1.000000	1.054200e+05
50%	2021.000000	5.000000	4.000000	2.000000	2.000000	1.000000	1.902120e+05
75%	2024.000000	14.000000	8.000000	3.000000	3.000000	2.000000	3.989300e+05
max	2316.000000	124.000000	202.000000	124.000000	2009.000000	43.000000	3.060283e+07

```
data.isnull().sum()
```

25%	2004.000000	2.000000	2.000000	-1.000000	-1.000000
50%	2021.000000	5.000000	4.000000	2.000000	1.000000
75%	2024.000000	14.000000	8.000000	2.000000	2.000000
max	2316.000000	124.000000	202.000000	3.000000	3.000000
			124.000000		2009.000000

```
data.isnull().sum()
```

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```

title          0
country        130
location       131
building_construction_year  82817
building_total_floors      79312
apartment_floor    92944
apartment_rooms    73358
apartment_bedrooms  110554
apartment_bathrooms  91563
apartment_total_area    5740
apartment_living_area  119824
price_in_USD      2575
image           0
url             0
dtype: int64

```

```
df= data.drop_duplicates()
```

```
print("Shape after removing duplicates:", df.shape)
```

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```
Shape after removing duplicates: (147536, 14)
```

Generate

+ Code

+ Markdown

```
plt.figure()

sns.histplot(df['price_in_USD'], bins=50)

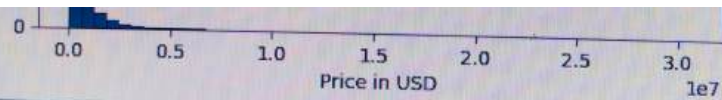
plt.title("Price Distribution")
plt.xlabel("Price in USD")
plt.ylabel("Count")

plt.show()
```

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...





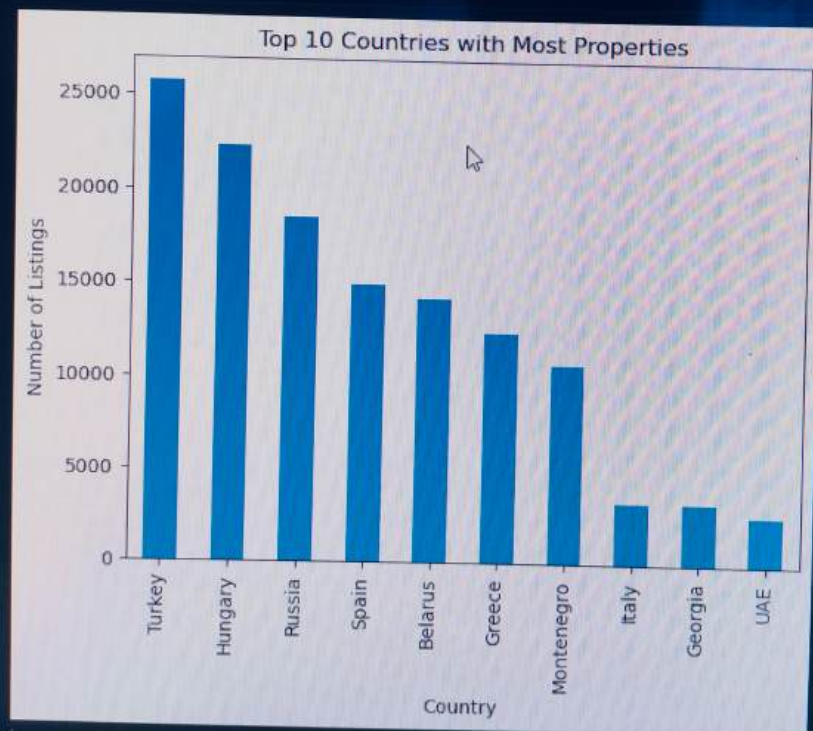
ate_data(147k).csv

ate_data.csv

```
plt.figure()
df['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Countries with Most Properties")
plt.xlabel("Country")
plt.ylabel("Number of Listings")
plt.show()
```

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...



te_data(147k).csv

te_data.csv

▼

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...

```
import matplotlib.pyplot as plt

plt.figure()
data['apartment_bedrooms'].value_counts().head(10).plot(kind='bar')
plt.title("Property Type Distribution")
plt.show()
```



estate_data.csv

estate_data(147k).csv

estate_data.csv

▼

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...

```
plt.figure()
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



```
import pandas as pd
```

```
# Create sample input (must match X.columns)
```

```
sample_house = pd.DataFrame([[2000, 3, 2, 5, 1, 0]], columns=X.columns)
```

```
# Predict price
```

```
predicted_price = model.predict(sample_house)
```

```
# Print labeled output
```

```
print("----- House Price Prediction -----")
```

```
print("Area (sqft):", sample_house.iloc[0,0])
```

```
print("Bedrooms:", sample_house.iloc[0,1])
```

```
print("Bathrooms:", sample_house.iloc[0,2])
```

```
print("Age of House:", sample_house.iloc[0,3])
```

```
print("Location Feature 1:", sample_house.iloc[0,4])
```

```
print("Location Feature 2:", sample_house.iloc[0,5])
```

```
print("-----")
```

```
print("Predicted House Price:", predicted_price[0])
```

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```
--- ----- House Price Prediction -----
```

```
Area (sqft): 2000
```

```
Bedrooms: 3
```

```
Bathrooms: 2
```

```
Age of House: 5
```

```
Location Feature 1: 1
```

```
Location Feature 2: 0
```

```
-----
```

```
Predicted House Price: 320092.6350238896
```